Analysis on the Use of AI among Students and Working Professionals

By Cheikh Diagne

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Author Note

GitHub Repository Link: https://github.com/cheikhdiagne2365-afk/AI-Use-Among-Students-and-Professionals

Original Database: https://www.kaggle.com/datasets/eishatuzzuhra/ai-usage-and-impact-on-students-and-professionals

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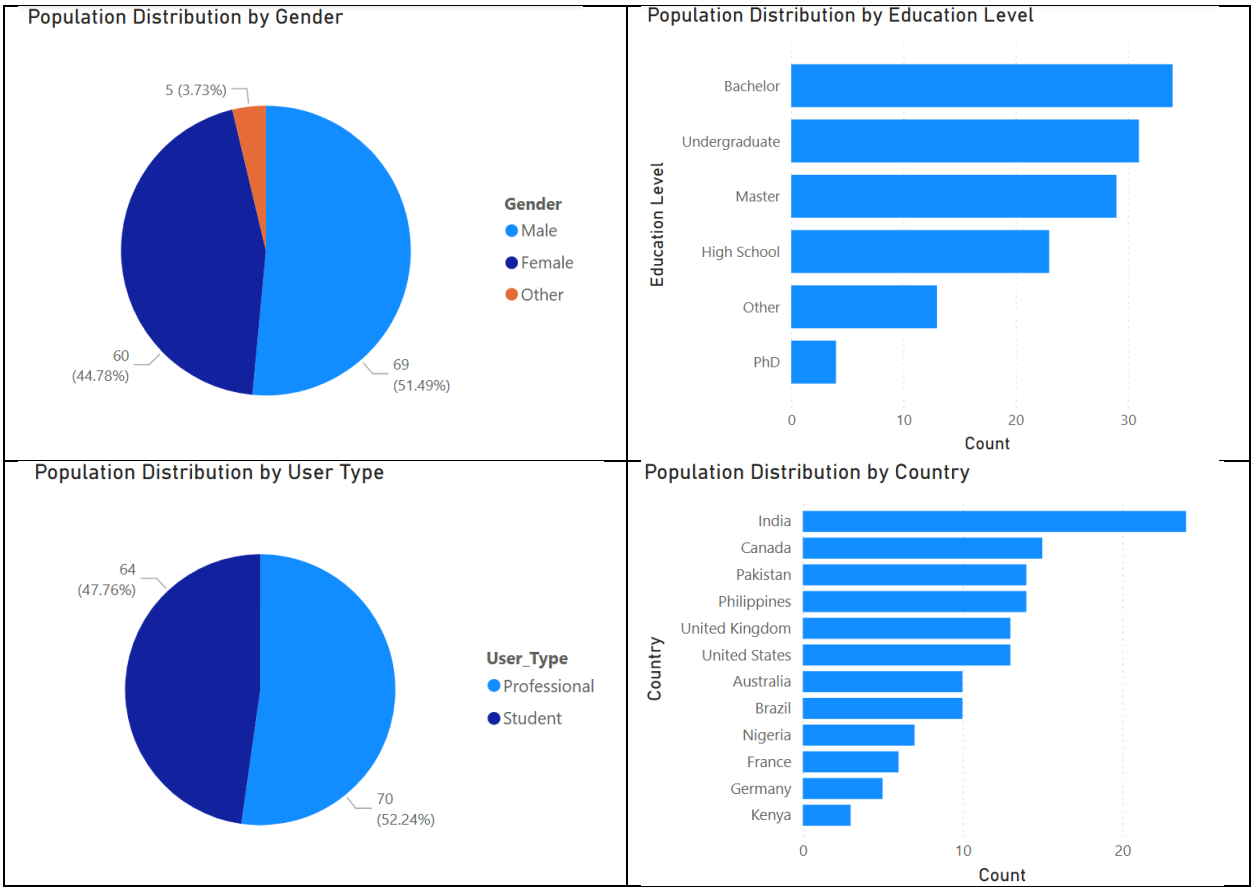
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Contextualization

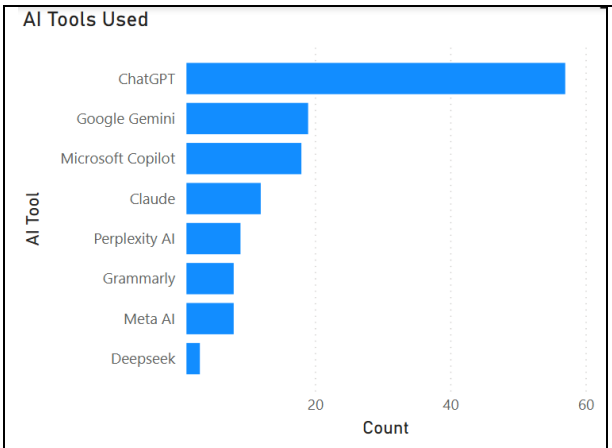
In today's modern landscape, the use of AI is becoming more normalized in people's day-to-day lives. It has slowly become a staple in the world of education and corporate society. Whether it is students using it to help with their schoolwork or professionals using it to optimize their daily tasks, it has shown to have some benefits. We're still in the infancy of AI and there are some drawbacks such as the accuracy of the provided information. This project aims to answer several questions: Is AI helping people be more productive? Are the tools satisfying the needs of the user? What tools lead to the best results? Do the pay-to-use AI tools lead to better results?

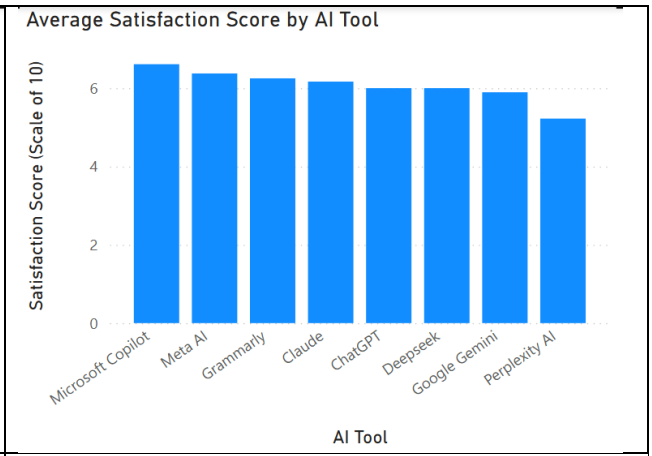
Population Analysis

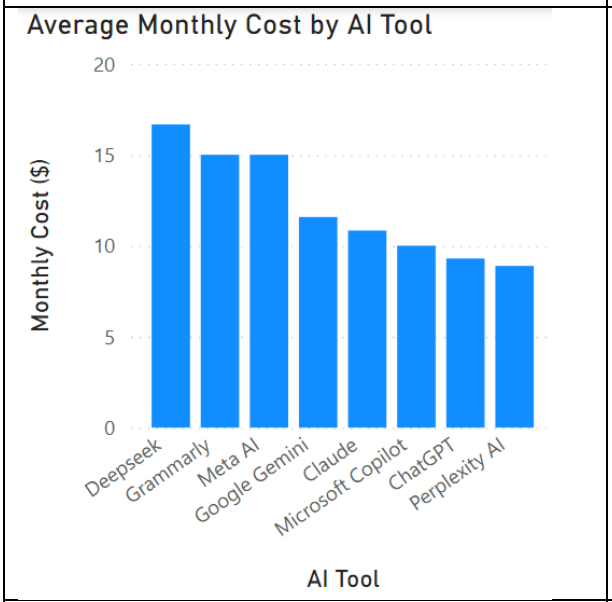
After performing the cleaning of the original database, we are left with 134 entries composed of students and working professionals. Of the 134, 70 identify themselves as professionals and 64 as students. There is an equal gender distribution with 69 people identifying as male, 60 identifying as female and 5 that identified as neither falling into the “Other” category. We notice an equal distribution of the education level in the observed population: Bachelor’s (34), Undergraduate (31), Master’s (29), High School (23), PhD (4), Other (13). For the countries that are represented, we notice a wide and diverse pool of individuals. India is the most represented country with 24 users followed by Canada with 15, Pakistan and the Philippines each with 14 and we have the United States and the UK rounding out the top 5 with 13.

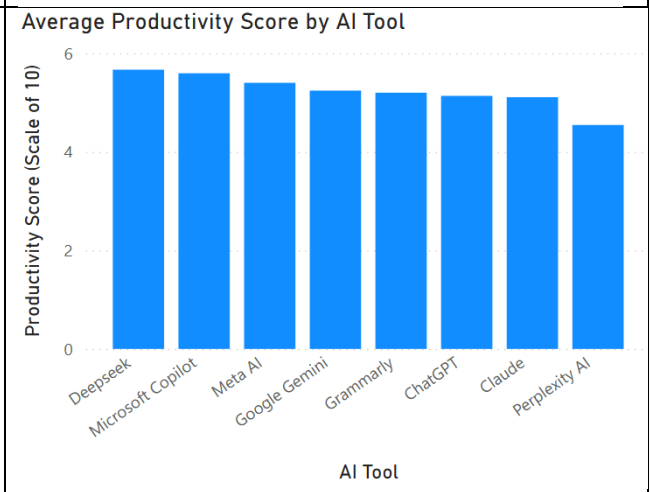


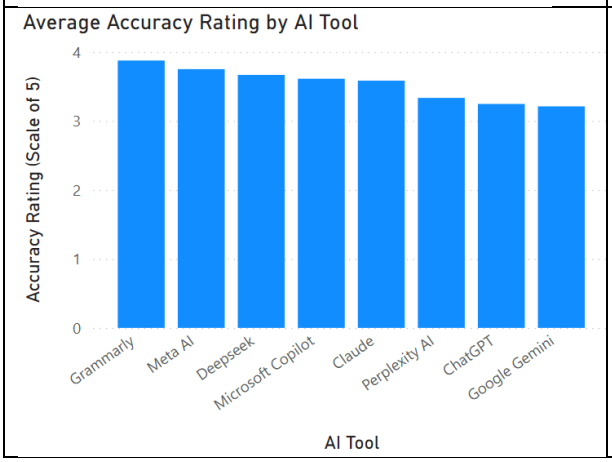
Analysis of the AI Tools Used











Satisfaction Score	
Tool	Score (Scale of 10)
ChatGPT	6
Google Gemini	5.89
Microsoft Copilot	6.61
Claude	6.17
Perplexity AI	5.22
Grammarly	6.25
Meta AI	6.38
Deepseek	6

Productivity Score	
Tool	Score (Scale of 10)
ChatGPT	5.14
Google Gemini	5.24
Microsoft Copilot	5.59
Claude	5.11
Perplexity AI	4.54
Grammarly	5.2
Meta AI	5.4
Deepseek	5.67

Tool Used	
Tool	Count
ChatGPT	57
Google Gemini	19
Microsoft Copilot	18
Claude	12
Perplexity AI	9
Grammarly	8
Meta AI	8
Deepseek	3

Monthly Cost	
Tool	Average Monthly Cost ($)
ChatGPT	9.3
Google Gemini	11.58
Microsoft Copilot	10
Claude	10.83
Perplexity AI	8.89
Grammarly	15
Meta AI	15
Deepseek	16.67

Accuracy Rating	
Tool	Rating (Scale of 5)
ChatGPT	3.25
Google Gemini	3.21
Microsoft Copilot	3.61
Claude	3.58
Perplexity AI	3.33
Grammarly	3.88
Meta AI	3.75
Deepseek	3.67

Relationship Analysis

For this section, we want to analyze the relationships between several different variables. This will allow us to answer some of our initial questions by determining if there's a correlation between the following variables:

- AI_Usage_Hours_Per_Day vs. Time_Saved_Hours_Per_Week
- AI_Usage_Hours_Per_Day vs. Productivity_Score
- Productivity_Score vs. Satisfaction_Score
- Time_Saved_Hours_Per_Week vs. Satisfaction_Score
- Monthly_AI_Cost vs. Accuracy_Rating
- Monthly_AI_Cost vs. Satisfaction_Score
- Monthly_AI_Cost vs. Productivity_Score

We will be doing a hypothesis test to determine if the correlation between the two variables is statistically significant. To determine this, we will need to calculate the p-value. The p-value is a number, that ranges from 0 to 1, that measures how well your data fits a "no effect" assumption, which is referred to as the null hypothesis. If the p-value is less than 0.05, then we can reject the null hypothesis and conclude that the correlation between variable A and B is statistically significant. The calculations were done using Microsoft Excel using all the ordered pairs in the scatter plots of the relationships and this process:

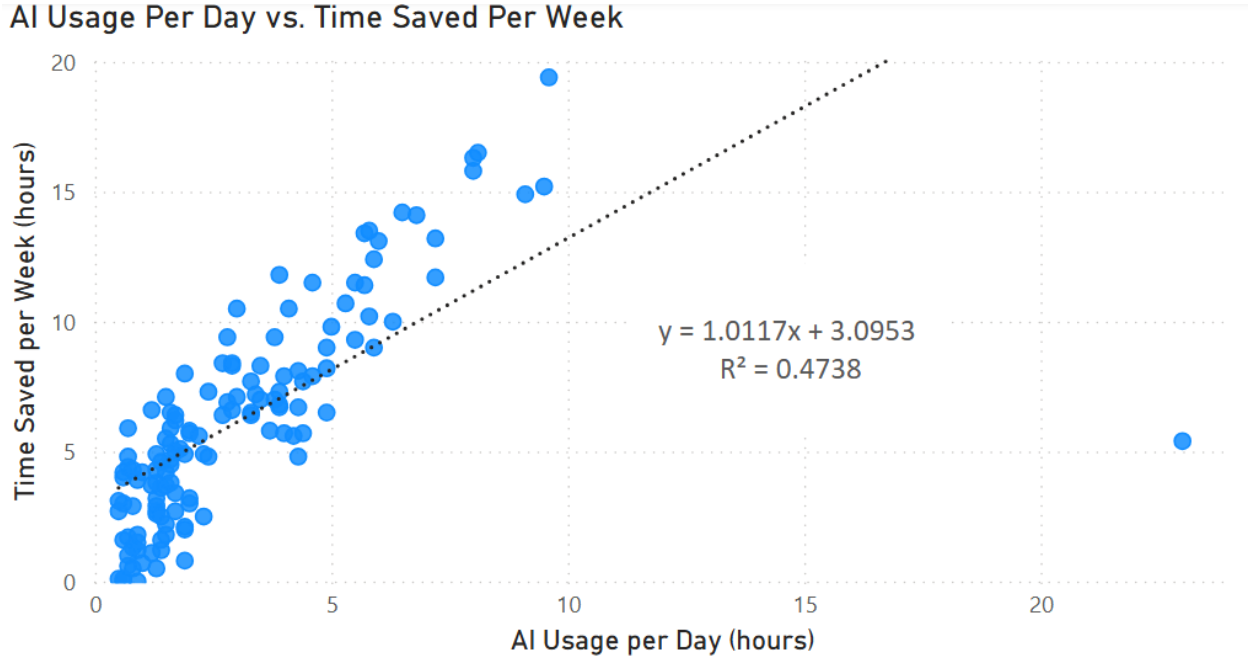
- Determine the R coefficient (correlation) $r = CORREL(array1, array2)$
- Calculate the t-statistic which measures how the sample data differs from a null hypothesis

$$t - statistic = r * \sqrt{\frac{n-2}{1-r^2}} \; where \; n \; is \; the \; number \; of \; ordered \; pairs \; (count)$$

- Calculate the p-value

$$p - value = T.DIST.2T(ABS(t - statistic), n - 2)$$

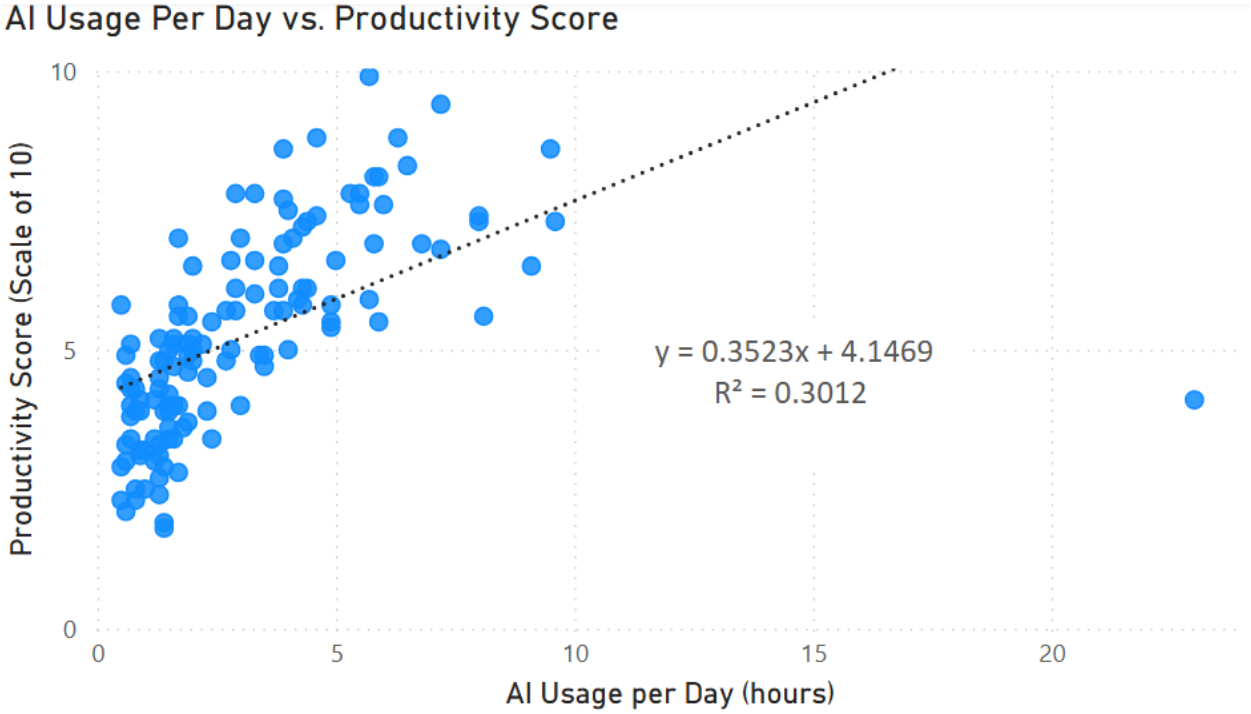
AI_Usage_Hours_Per_Day vs. Time_Saved_Hours_Per_Week



R coefficient	0.68834318
n	133
t-statistic	10.8610542
p-value	5.44E-20

Analysis: The p-value for this relationship is 5.44E-20. Since the p-value is lower than 0.05, we can conclude that the relationship between the daily AI usage (hours) and time saved per week (hours) is statistically significant meaning that it is not due to chance. The linear regression for this relationship yields a slope of $y = 1.0117x + 3.0953$. This means that for every hour of AI usage, the user saves an additional 1.0117 hour per week. The coefficient of determination (R^2) is 0.4738. This means that 47.38% of the variation of y (Time Saved per Week) is accounted for by x (AI Usage per Day).

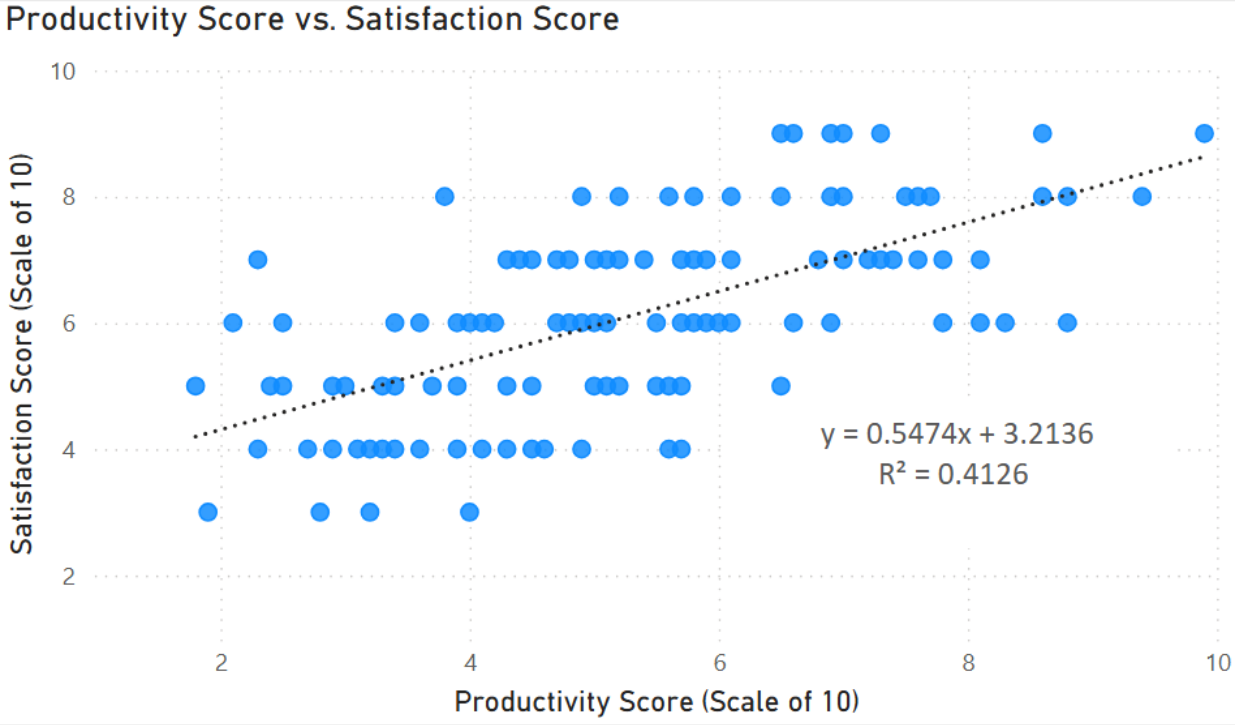
AI_Usage_Hours_Per_Day vs. Productivity_Score



R coefficient	0.5488146
n	130
t-statistic	7.42768381
p-value	1.38E-11

Analysis: The p-value for this relationship is 1.38E-11. Since the p-value is lower than 0.05, we can conclude that the relationship between the daily AI usage (hours) and the productivity score is statistically significant meaning that it is not due to chance. The linear regression for this relationship yields a slope of $y = 0.3523x + 4.1469$. This means that for every hour of AI usage, the average productivity score increases by 0.3523. The coefficient of determination (R^2) is 0.3012. This means that 30.12% of the variation of y (Productivity Score) is accounted for by x (AI Usage per Day).

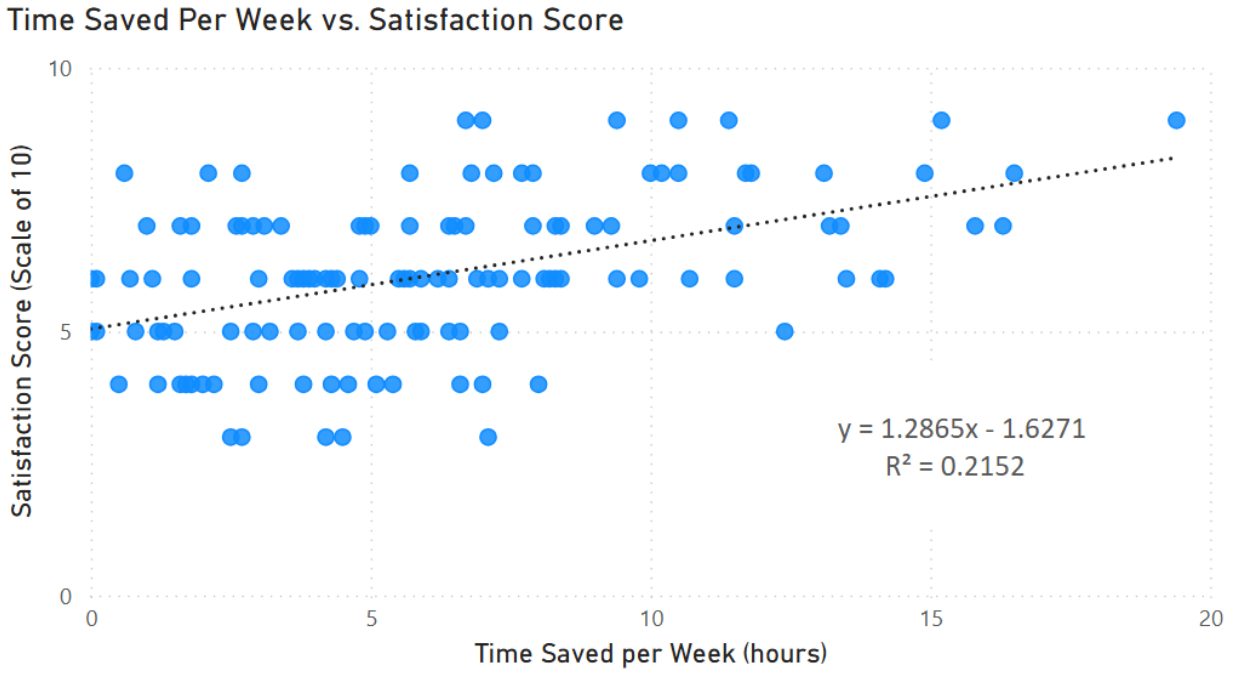
Productivity_Score vs. Satisfaction_Score



R coefficient	0.64230818
n	107
t-statistic	8.58729278
p-value	8.86E-14

Analysis: The p-value for this relationship is 8.86E-14. Since the p-value is lower than 0.05, we can conclude that the relationship between productivity score and satisfaction score is statistically significant meaning that it is not due to chance. The linear regression for this relationship yields a slope of $y = 0.5474x + 3.2136$. This means that for every increase of 1 of the productivity score the satisfaction score increases by 0.5474. The coefficient of determination (R^2) is 0.4126. This means that 41.26% of the variation of y (Satisfaction Score) is accounted for by x (Productivity Score).

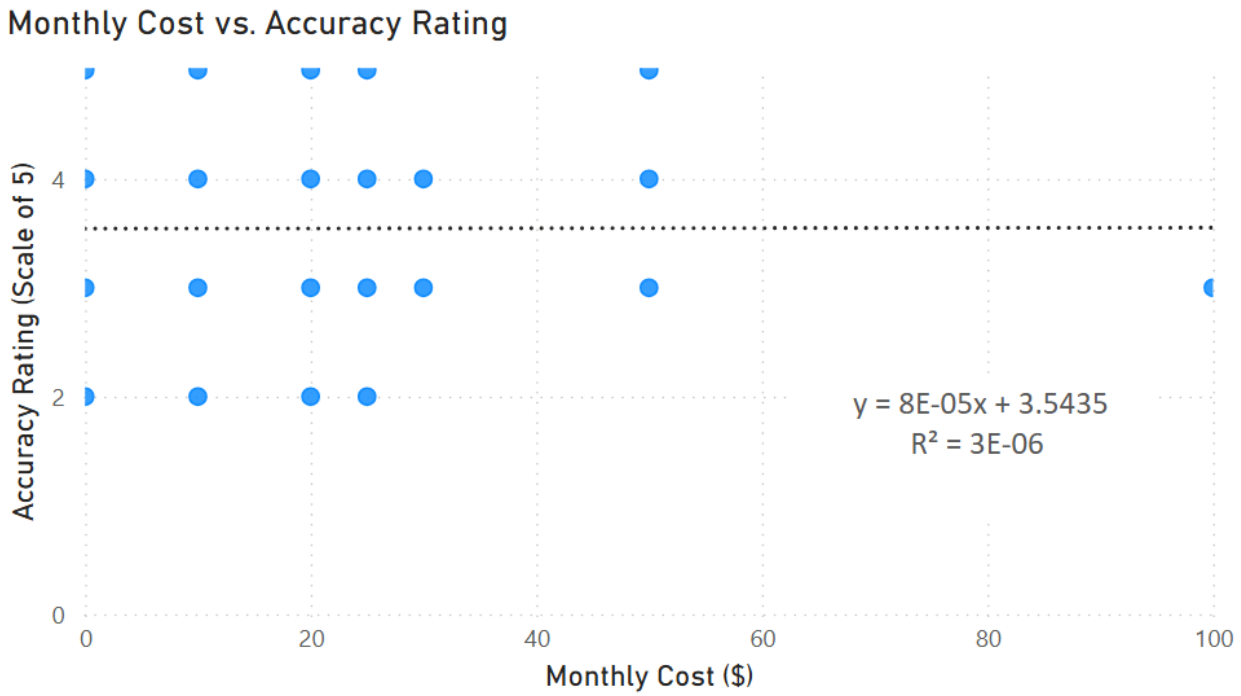
Time_Saved_Hours_Per_Week vs. Satisfaction_Score



R coefficient	0.46384428
n	125
t-statistic	5.80673293
p-value	5.10E-08

Analysis: The p-value for this relationship is 5.10E-08. Since the p-value is lower than 0.05, we can conclude that the relationship between the time saved per week (hours), and the satisfaction score is statistically significant meaning that it is not due to chance. The linear regression for this relationship yields a slope of $y = 1.2865x - 1.6271$. This means that for every hour saved per week, the satisfaction score increases by 1.2865. The coefficient of determination (R^2) is 0.2152. This means that 21.52% of the variation of y (Satisfaction Score) is accounted for by x (Time Saved per Week).

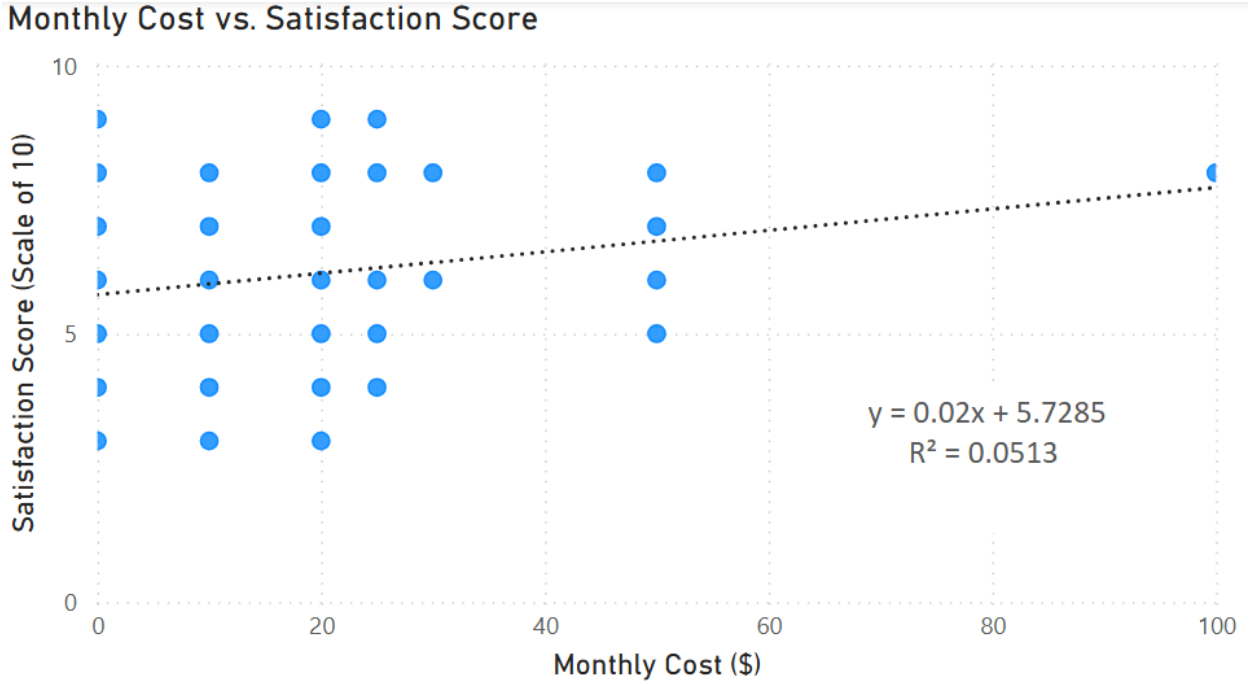
Monthly_AI_Cost vs. Accuracy_Rating



R coefficient	0.00178719
n	22
t-statistic	0.00799256
p-value	0.994

Analysis: The p-value for this relationship is 0.994. Since the p-value is higher than 0.05, we can conclude that the relationship between monthly cost ($) and the accuracy rating is not statistically significant meaning that it is due to chance.

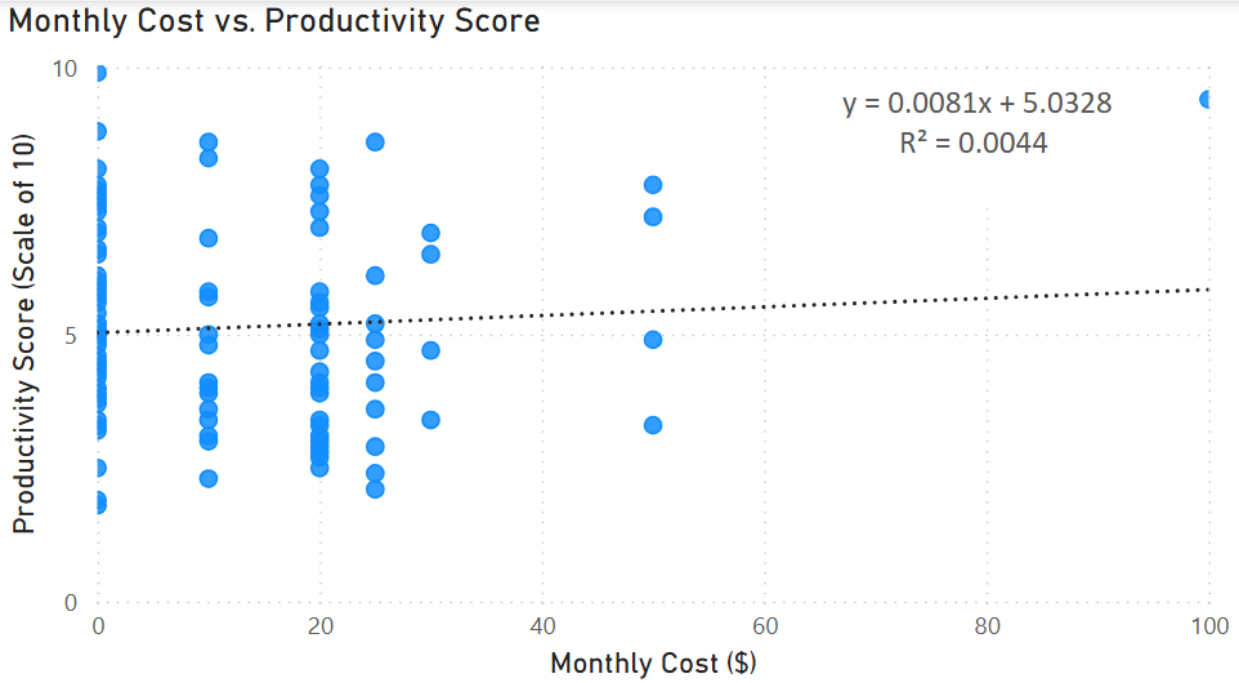
Monthly_AI_Cost vs. Satisfaction_Score



R coefficient	0.22659988
n	32
t-statistic	1.27428543
p-value	0.212

Analysis: The p-value for this relationship is 0.212. Since the p-value is higher than 0.05, we can conclude that the relationship between monthly cost ($) and the satisfaction score is not statistically significant meaning that it is due to chance.

Monthly_AI_Cost vs. Productivity_Score



R coefficient	0.06661411
n	98
t-statistic	0.65413528
p-value	0.515

Analysis: The p-value for this relationship is 0.515. Since the p-value is higher than 0.05, we can conclude that the relationship between monthly cost ($) and the productivity score is not statistically significant meaning that it is due to chance.

Conclusion

At the beginning of this project, our initial questions were: Is AI helping people be more productive? Are the tools satisfying the needs of the user? What tools lead to the best results? Do the pay-to-use AI tools lead to better results? To begin, the analysis of the relationships involving the daily use of AI (AI_Usage_Hours_Per_Day) allowed us to conclude that an increase in daily AI usage leads to the user saving more time over the week. It also leads to an increase to the user's productivity score. Both relationships had a p-value below the mark of 0.05, meaning that their relationship between the variables was statistically significant. In the end, we can conclude that AI is helping people optimize their time and be more productive. The same outcome can be observed for both relationships involving the user's satisfaction of their AI use (Satisfaction_Score). Both variables (Productivity_Score & Time_Saved_Hours_Per_Week) are shown to have statistically significant relationships with the user's satisfaction score. Each relationship shows that there an increase of the dependant variable (Satisfaction_Score), when there's an increase of the independent variable (Productivity_Score & Time_Saved_Hours_Per_Week). We can conclude that the use of AI is satisfying the users. The database has a total of 8 different AI platforms that were being used. We notice a common trend in the 3 key performance indicators (Satisfaction_Score, Accuracy_Rating, Productivity_Score) for all the tools. For the Satisfaction_Score & Accuracy_Rating, which are rated on a scale of 10, the former has ratings that range between 5 and 7 while the latter's ratings range between 4.5 and 6. Both variables have low standard deviations (Satisfaction_Score: 0.413106697, Productivity_Score: 0.34875032), which allows us to conclude that there isn't much of a difference in rating between the different AI tools. Microsoft Copilot has the highest satisfaction score at 6.61, while the lowest is Perplexity AI at 5.22. Deepseek has the highest productivity score at 5.67 and the lowest is, once again, Perplexity AI at 4.54. We notice that Perplexity AI distinguishes itself from the pack as the least effective AI Tool, in the eyes of the users. If we were to exclude Perplexity AI, there is a 39% and 36% drop-off in the standard deviation for the satisfaction score and productivity score respectively. We notice the same for the accuracy rating; all the ratings vary between 3 and 4 on a scale of 5. The standard deviation for the accuracy rating is 0.224948974. In the end, there isn't one tool that separates itself from the others in terms of performance and the results it provides to the users. We performed a relationship analysis for the monthly cost variable against all 3 key performance indicators. It showed that the monthly cost of on AI tool doesn't have a statistically significant relationship with the satisfaction score, productivity score or accuracy rating. We can conclude, that paying to use an AI tool will not necessarily lead to better or more accurate results for the user.

Annex

Annex 1: AI_Tool vs. Would_Recommend

AI_Tool	Maybe	No	Yes	**Total**
ChatGPT	8	20	29	**57**
Claude	4	2	6	**12**
Copilot	1		1	**2**
Deepseek		1	2	**3**
Google Gemini	1	3	15	**19**
Grammarly		2	6	**8**
Meta AI	2	2	4	**8**
Microsoft Copilot	1	5	10	**16**
Perplexity AI	2	4	3	**9**
Total	**19**	**39**	**76**	**134**

Annex 2: AI_Purpose vs. Would_Recommend

AI_Purpose ▲	Maybe	No	Yes	**Total**
Coding	5	11	23	**39**
Content Creation	2	3	8	**13**
Customer Support	3	2	5	**10**
Data Analysis	2	3	5	**10**
Office Work	1		2	**3**
Research	2	9	11	**22**
Studying	1	7	7	**15**
Writing	3	4	15	**22**
Total	**19**	**39**	**76**	**134**

Annex 3: User_Type vs. Would_Recommend

User_Type	Maybe	No	Yes	**Total**
Professional	13	17	40	**70**
Student	6	22	36	**64**
Total	**19**	**39**	**76**	**134**

Annex 4: AI_Tool vs. AI_Purpose

AI_Tool	Coding	Content Creation	Customer Support	Data Analysis	Office Work	Research	Studying	Writing	**Total**
ChatGPT	15	6	4	5	1	7	6	13	**57**
Claude	1	4		2		2	1	2	**12**
Copilot	1					1			**2**
Deepseek	1		1			1			**3**
Google Gemini	8	1	2	1	2	2	2	1	**19**
Grammarly	2					2	2	2	**8**
Meta AI	2	1	1			2	1	1	**8**
Microsoft Copilot	7	1		1		3	2	2	**16**
Perplexity AI	2		2	1		2	1	1	**9**
Total	**39**	**13**	**10**	**10**	**3**	**22**	**15**	**22**	**134**

Annex 5: AI_Tool vs. Education_Level

AI_Tool	Bachelor	High School	Master	Other	PhD	Undergraduate	**Total**
ChatGPT	16	6	14	5	2	14	**57**
Claude	1	6	1			4	**12**
Copilot				1		1	**2**
Deepseek		1		1		1	**3**
Google Gemini	6	3	5	3		2	**19**
Grammarly	2			1	2	3	**8**
Meta AI	2	2	2	1		1	**8**
Microsoft Copilot	5	3	4	1		3	**16**
Perplexity AI	2	2	3			2	**9**
Total	**34**	**23**	**29**	**13**	**4**	**31**	**134**

Annex 6: User_Type vs. AI_Purpose

User_Type	Coding	Content Creation	Customer Support	Data Analysis	Office Work	Research	Studying	Writing	**Total**
Professional	26	7	10	7	3	9		8	**70**
Student	13	6		3		13	15	14	**64**
Total	**39**	**13**	**10**	**10**	**3**	**22**	**15**	**22**	**134**